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# WAVELET ANALYSIS OF EEG SIGNALS DURING MOTOR IMAGERY

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**Abstract:**

**ERS and ERD (event-related synchronization and desynchronization) are observed in EEG (electroencephalogram) signals around such events as sensitive stimulus, motions, cognitive actions etc. Usually, ERS/ERD features of EEG are extracted as variances of band-passed signals of several trials. To make use of these features to recognize inputs for BCI (brain-computer interface), we applied discrete wavelet analysis to extraction of ERS/ERD features from a small number of EEG signals during motor imagery. We employed Daubechies, convolution and spline biorthogonal mothers for linear wavelet analysis, also Haar type structural function for morphological wavelet analysis. Then our extraction method was estimated by the pattern recognition based on AR model.**

**Keywords:**

**EEG; BCI; ERS; ERD; Motor imagery; Multiresolution analysis; Mathematical morphology; Intertrial variance**

## 1. Introduction

Electroencephalograph (EEG) signals can be used to move a cursor to a target on a computer screen. Such an EEG-based brain computer interface (BCI) can provide a new communication channel to replace an impaired motor function ([1], [2]). It can be used by e.g., handicapped users with amyotrophic lateral sclerosis (ALS). In such a BCI system, information (e.g. ERD/ERS, P300 etc.) which is included in EEG signals related to a cognitive task or motor imagery is used in order to estimate the will of humans.

For the EEG signals recognition method, there are many methods have been proposed. For example, Pfurtscheller et al extracted ERD/ERS from EEG during right and left hand motor imagery and made it use for the pattern recognition for single-trial wave form by a neural network [3]. Guger et al developed a spelling device using P300 feature [4]. We studied pattern recognition method based on autoregressive (AR) model about right and left hand motor imagery [5]. However, by this technique, we could not make it clear significant feature for the pattern identification in this technique.

In this work, we will extract a significant component for the pattern identification from right and left hand motor imagery. These components appear in various frequency bands and these dominant bands usually depend on subject. Therefore, we will identify significant frequency bands by using multiresolution analysis in order to extract these components. In practice, by applying wavelet analysis [6] and mathematical morphological multiresolution analysis ([7], [8]), we extract changes of power caused by motor imagery from EEG signal in various frequency components.

## 2. Feature extraction and pattern recognition

### 2.1. Pattern recognition method based on AR model

EEG signals are assumed to be generated from an autoregressive (AR) model. Namely, let $y_t$ be the observed EEG signal at time $t$ and $\nu_t$ be an independent random variable with the normal distribution

$$E[\nu_t \nu_s] = \rho\delta_{t,s}, \quad \nu_t \sim N[0,\rho].$$

Then the behavior of EEG is expressed as

$$y_t = \sum_{j=1}^{m} \phi_j y_{t-j} + \nu_t = \Phi^T \boldsymbol{z}_{t-1} + \nu_t \qquad (m+1 \le t \le N) \tag{2.2.1}$$

with the parameter vectors $\Phi = [\phi_1, \phi_2, \ldots, \phi_m]^T$ and $\boldsymbol{z}_{t-1} = [y_{t-1}, y_{t-2}, \ldots, y_{t-m}]^T$. The first $m$ observations $y_1, y_2, \ldots, y_m$ serve as the initial conditions for the Eq. (2.2.1). We also combine the parameters as a single vector $\theta = \left[\Phi^T, \rho\right]^T$ and refer it as the feature parameter.

For the feature parameter $\theta_k$ corresponding to the class $\omega_k$, we make assumptions of independency and determinacy. Namely, we assume it satisfies the independency condition

$$p(y_1, y_2, \ldots, y_m, \theta_k) = p(y_1, y_2, \ldots, y_m)\, p(\theta_k) \tag{2.2.2}$$

and the determinacy condition

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$$p(\boldsymbol{y}_N | \omega_k) = p(\boldsymbol{y}_N | \theta_k) \quad (2.2.3)$$

are satisfied. Here $\boldsymbol{y}_N = [y_1, y_2, \ldots, y_N]^T$.

These assumptions yield the following explicit expression for the conditional probability density functions.

$$p(\boldsymbol{y}_N | \omega_k) = p(y_1, y_2, \ldots, y_m)\left(\frac{1}{2\pi\rho_k}\right)^{\frac{N-m}{2}} \cdot \exp\left[-\frac{1}{2\rho_k}\sum_{t=m+1}^{N}\left(y_t - \Phi_k^T z_{t-1}\right)^2\right] \quad (2.2.4)$$

The aim of this work is to identify two classes of human's will. We adopt the following Bayesian decision rule.

$$\begin{aligned} k^* &= Arg\,\underset{k}{Max}\,\Pr(\omega_k | \boldsymbol{y}_N) \\ &= Arg\,\underset{k}{Max}\left\{-\frac{N-m}{2}\ln(2\pi\rho_k) \right. \quad (2.2.5) \\ &\left. -\frac{1}{2\rho_k}\sum_{t=m+1}^{N}\left(y_t - \Phi_k^T \boldsymbol{z}_{t-1}\right)^2 + \ln\Pr(\omega_k)\right\} \end{aligned}$$

where $\boldsymbol{y}_N = [y_1, y_2, \ldots, y_N]^T$ is a time sequence of EEG signal and $\omega_k$ represents the $k$-th class. To estimate these parameters, the likelihood function

$$L = \sum_{j=1}^{Np} p\left(\boldsymbol{y}_N^j | \omega_k\right) \quad (2.2.6)$$

is adopted as the criterion. Where $N_P$ is the number of samples of given class $k$ and $\boldsymbol{y}_N^j$ and $\boldsymbol{z}_{t-1}^j$ are $j$-th sample. To maximize $L$, put all of the 1-st order partial differentials of $L$ to be zero. Then the derived simultaneous equations are solved to give rise to the following relations.

$$\hat{\Phi}_k = \left[\sum_{j=1}^{Np}\sum_{t=m+1}^{N} \boldsymbol{z}_{t-1}^j \boldsymbol{z}_{t-1}^{jT}\right]\cdot\left[\sum_{j=1}^{Np}\sum_{t=m+1}^{N} y_t^j \boldsymbol{z}_{t-1}^{jT}\right], \quad (2.2.7)$$

$$\hat{\rho}_k = \frac{1}{Np(N-1)}\sum_{j=1}^{Np}\sum_{t=m+1}^{N}\left(y_t^j - \hat{\Phi}_t^T \boldsymbol{z}_{t-1}^j\right)^2. \quad (2.2.8)$$

## 2.2. Multiresolution analysis

In this work, we use both wavelet and mathematical morphological multiresolution analysis. These are captured as a unified manner [8]. Since one can find introductory description about wavelet multiresolution in many literatures (e.g. [6]), we give a brief description only on mathematical morphological multiresolution analysis. For a precise description, see [8].

As basic operations, we employ Minkowski addition $\oplus$ and Minkowski subtraction $\ominus$, those are respectively defined as follows.

$$[f \oplus g](t) = \max_{\substack{t-u\in F \\ u\in G}}\{f(t-u) + g(u)\} \quad (2.3.1)$$

$$[f \ominus g](t) = \min_{u\in G}\{f(t-u) - g(u)\} \quad (2.3.2)$$

Here $f(t)$ is the input signal and $g(t)$ is the structural function which characterizes the filters which will be constructed below. The sets $F$ and $G$ denote the domains of $f(t)$ and $g(t)$, respectively. Conventionally, it is assumed that every signal takes the value $-\infty$ out of its domain.

By combining these operations, we define two morphological filters:

Opening: $f_g(t) = [(f \ominus g^s) \oplus g](t)$, (2.3.3)

Closing: $f^g(t) = [(f \oplus g^s) \ominus g](t)$, (2.3.4)

where $g^s(t) := g(-t)$. The opening process for $f(t)$ by $g(t)$ removes the parts in the positive direction of the wave form of $f(t)$ those are too narrow to fit for the wave form of $g(t)$ attached below. In contrast, the closing filter removes the narrow negatively directed parts. In other words, opening (resp. closing) smooth $f(t)$ from the positive (resp. negative) direction by $g(t)$. Furthermore, the open-closing filter consisting of successive applications of the opening followed by the closing provides an effect of low pass filter. Thus, we can also construct a high pass filter by taking the difference between the input signal and its open-closed result.

Low pass: $\psi^{\uparrow}(t) = (f_g)^g(t)$ (2.3.5)

High pass: $\omega^{\uparrow}(t) = f(t) - \psi^{\uparrow}(t)$ (2.3.6)

Notations are borrowed from [8]. The up arrow indicates increasing of the level in multiresolution. The schematic figures of morphological operations are shown in Figure 2.1

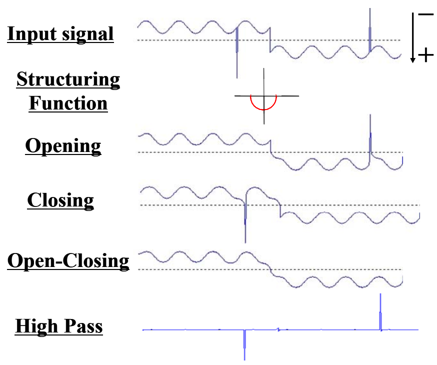


Figure. 2.1 Property of morphological filter

It is necessary to set out the structural function to use morphological filters. By the convention, the structural function has finite values only in the processing window and takes $-\infty$ on the outside. In this paper, we use the Haar type structural functions. The Haar type structural function is defined with a constant parameter $c$ and the width of window $2n+1$ as follows.

$$g(t) = \begin{cases} c & (-n \le t \le n) \\ -\infty & (otherwise) \end{cases} \tag{2.3.7}$$

A sequence of successive processes with varying structural functions constitutes a multiresolution signal analysis. To describe this more precisely, let us assume that there exist sets $V_j$ and $W_j$ for each level $j$ ($j = 0$, 1, …). We refer to $V_j$ (resp. $W_j$) as the signal space (resp. detail space) at the level $j$. Then, for a given input signal $x_0 \in V_0$, we obtain the following recursive analysis scheme.

$$\begin{aligned} x_0 &\to \{x_1, y_1\} \to \{x_2, y_2, y_1\} \to \cdots \\ &\to \{x_k, y_k, y_{k-1}, \ldots, y_1\} \to \cdots \end{aligned} \tag{2.3.8}$$

where

$$x_{j+1} = \psi_j^{\uparrow}(x_j) \in V_{j+1}, \quad y_{j+1} = x_j - x_{j+1} \in W_{j+1}.$$

for $x_j \in V_j$, $x_{j+1} \in V_{j+1}$.

Conversely, the input signal $x_0$ can be reconstructed by summing up the detail signals at every level. A three-level signal analysis scheme is depicted in Fig. 2.2. In this work, we increase the width of windows by 2 to the power of levels.

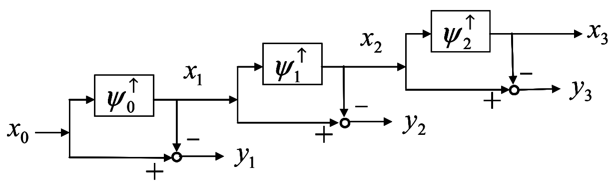


Figure 2.2 Three-level signal analysis scheme

### 2.3. Pattern recognition method based on Intertrial Variance component

Several methods for extraction of ERD/ERS features are proposed. In this article, we employ the Intertrial Variance (IV) method that is rather easily obtained.

#### 2.3.1. Notations

We summarize the notations used in this section. In what follows, the discrete time parameter $t$ runs from 1 through $N$.

$M$: number of trials

$x(t)$: EEG signal

$x^i(t)$: EEG signal of the $i$-th trial

$y(t)$: bandpass filtered signal for $x(t)$

$y^i(t)$: bandpass filtered signal for $x^i(t)$

$\boldsymbol{y} = [y(1), y(2), \ldots, y(N)]^T$

$\boldsymbol{y}^i = [y^i(1), y^i(2), \ldots, y^i(N)]^T$

$\boldsymbol{y}_k^i = [y_k^i(1), y_k^i(2), \ldots, y_k^i(N)]^T$ $\boldsymbol{y}^i$ of the class $k$

$\overline{\boldsymbol{y}} = [\overline{y}(1), \overline{y}(2), \ldots, \overline{y}(N)]^T$ intertrial average of $\boldsymbol{y}$

#### 2.3.2. IV method

Intertrial Variance is defined by

$$\begin{cases} IV(t) = \dfrac{1}{M-1} \displaystyle\sum_{i=1}^{M} \left( y^i(t) - \overline{y}(t) \right)^2 \\ \overline{y}(t) = \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} y^i(t) \end{cases} \tag{2.3.1}$$

Thus $IV$ measures the mean square distance between the signal and the intertrial average. The ERD/ERS features are obtained as positive/negative peaks of the ratio of the IV value to the mean value at rest.

#### 2.3.3. Probability density function $p(\boldsymbol{y}|\omega_k)$

By assuming that the mean square distance between the signal and the intertrial average subordinate to normal distribution, the probability density function $p(\boldsymbol{y}|\omega_k)$ can be explicitly represented as follows:

$$\begin{aligned} p(\boldsymbol{y}|\omega_k) &= p(\boldsymbol{y}|\theta_k) \\ &= \frac{1}{(2\pi)^{\frac{N}{2}} |\Sigma_k|^{\frac{1}{2}}} \exp\left\{ -\frac{1}{2} (\boldsymbol{y} - \overline{\boldsymbol{y}}_k)^T \Sigma_k^{-1} (\boldsymbol{y} - \overline{\boldsymbol{y}}_k) \right\} \end{aligned} \tag{2.3.2}$$

$$\theta_k := (\overline{\boldsymbol{y}}_k, \Sigma_k) \tag{2.3.3}$$

Here

$$\overline{\boldsymbol{y}}_k = E\{\boldsymbol{y}_k\},$$

$$\Sigma_k = E\left\{ (\boldsymbol{y} - \overline{\boldsymbol{y}}_k)(\boldsymbol{y} - \overline{\boldsymbol{y}}_k)^T \right\}.$$

Furthermore, we assume that the difference from averaging vector is componentwise independent. That is, the covariance matrix $\Sigma_k$ is diagonal:

$$\Sigma_k = diag\left[\sigma_{k,1}^2, \sigma_{k,2}^2, \ldots, \sigma_{k,N}^2\right],$$
$$\sigma_{k,t}^2 = E\left\{\left(y_k(t) - \overline{y}_k(t)\right)^2\right\}$$
$$(t = 1,2,\ldots,N).$$

### 2.3.4. Estimation of density functions

The estimation of the density function $p(\boldsymbol{y}|\omega_k)$ for the class $k$ is reduced to those for the parameters $\overline{\boldsymbol{y}}_k$ and $\sigma_{k,t}^2$ $(t = 1,2,\ldots,N)$. We assume that these are estimated as follows:

$$\overline{\boldsymbol{y}}_k = \frac{1}{M}\sum_{i=1}^{M}\boldsymbol{y}_k^i,$$
$$\sigma_{k,t}^2 = \frac{1}{M-1}\sum_{i=1}^{M}\left(y_k^i(t) - \overline{y}_k(t)\right)^2$$
$$(t = 1,2,\ldots,N).$$

### 2.3.4. Pattern recognition method

As the pattern recognition method, we adopt Bayesian discrimination. Then by using (2.3.2), our discrimination function becomes

$$k^* = Arg\,\underset{k}{Max}\,\Pr(\omega_k|\boldsymbol{y})$$
$$= Arg\,\underset{k}{Min}\left\{\frac{1}{2}\sum_{t=1}^{N}\ln\sigma_{k,t}^2 + \frac{1}{2}\sum_{t=1}^{N}\left\{\frac{1}{\sigma_{k,t}^2}\left(y(t) - \overline{\boldsymbol{y}}_k(t)\right)^2\right\} - \ln\Pr(\omega_k)\right\} \tag{2.3.4}$$

Since the constant term has no effect on the result, we have (2.3.4).

## 3. Method and materials

### 3.1. Experimental Paradigm

For the right and left hand motor imagery task, the response brain waves caused by hand motor imagination is observed. In this study, three subjects (A, B, C) participated in the experiment performed following the experimental paradigm below.

Each trial takes 8000 ms long (Figure. 3.1). At 3000 ms, the fixation cross is overlaid with an arrow at the center of the monitor for 1250 ms, pointing either to the left (←) or to the right (→). According to the direction of the arrow, the subject is instructed to imagine a movement of the left hand or the right hand. Feedback (FB) consisting of a bar-graph is indicated at the center of the monitor from 5000 ms to 8000 ms. Subjects participate in 10 sessions all on different days. Each session consists of 3 experimental runs of 60 trials (30 'left' and 30 'right') and lasts about 40 minutes. By ranging the length of breaks between trials from 500 ms to 2500 ms, the occurrences of trials in each run are randomized.

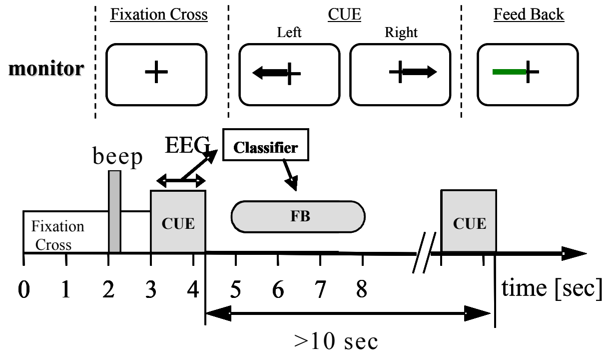


Figure 3.1 Timing chart

There are two types of sessions: in the initial session, data is collected to create a subject-specific classifier and, hence, no feedback is provided. In the following sessions, the classifier is used to classify the subject's EEG on-line while he imagines the movement requested, and feedback is given to the subject as described above. From the second session, feedbacks based on the parameters estimated by using the data of the previous session are executed.

### 3.2. EEG Recording and Data Acquisition

EEG recording and data acquisition are done as follows. EEG recording electrode positions are shown in Figure 3.2. For example, channel C3 is derived by following equation.

Monopolar: $y_t = y_{\mathrm{C3}} - y_{\mathrm{A2}}$

Bipolar: $y_t = y_{\mathrm{C3A}} - y_{\mathrm{C3P}}$

Small Laplacian: $y_t = y_{\mathrm{C3}} - \frac{y_{\mathrm{C3A}} + y_{\mathrm{C3R}} + y_{\mathrm{C3P}} + y_{\mathrm{C3L}}}{4}$

where the positions of electrodes $y$ 's are as follows

$y_{\mathrm{C3}}$: at C3,
$y_{\mathrm{C3A}}$: at 2.5 cm anterior to C3,
$y_{\mathrm{C3R}}$: at 2.5 cm right to C3,
$y_{\mathrm{C3P}}$: at 2.5 cm posterior to C3,
$y_{\mathrm{C3L}}$: at 2.5 cm left to C3,
$y_{\mathrm{A2}}$: on the right earlobe.

The channel C4 is derived in the same way. The EEG signals are amplified and band-pass filtered between 1.5 and

60 Hz and then sampled at 500 Hz.

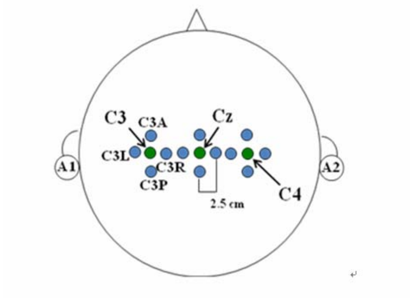


Figure 3.2 Positions of the electrodes

## 4. Result

Each EEG signal was decomposed into 8 levels by wavelet analysis. Mothers wavelets we used are the following 14 types: Daubechies ($N$ = 2, 4, 6, 8), Convolution types (1, 2, 3, 4) and Splines (B, O and C-spline functions, degrees 1 and 3 for each). An example of decomposition result is shown in Figure 4.1.

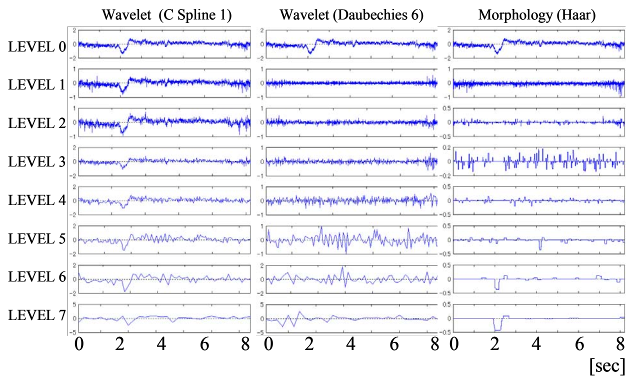


Figure 4.1 Multiresolution analysis (channel: C3, subj.: A, session : 10, class: left, trial : 14)

By using mathematical morphological method, peaks appearing at 2 sec and 4 sec are observed in less numbers of bandwidths (levels 5-7). Fig. 4.2. shows the IV obtained by (2.4.1) for the wave forms decomposed by Daubechies type mother wavelets, In particular, we remark that in levels 1-4, a decrease of power (ERD) is observed at the initial time (3 sec) of the motor imagery and a slow increase of power (ERS) is observed until the stopping time (8 sec).

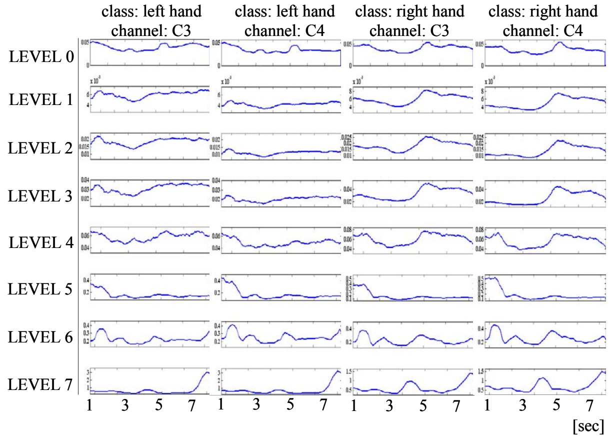


Figure 4.2 Intertrial Variance (process: Wavelet, mother: Daubechies 6, subj.: A, session: 5, smoothed)

There is no significant difference of IV among channels but differences in powers are observed among classes. The result for pattern recognition by using (2.3.2) is shown in Figure 4.3.

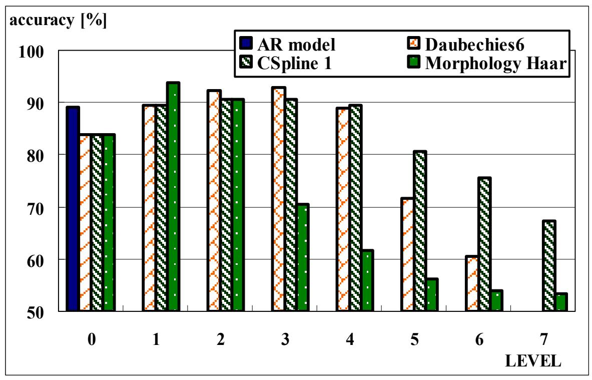


Figure 4.3 Pattern classification using IV(channel: C3, Subj.: A, session: 10)

To obtain a more general view of the ability of classification, a 10 fold cross validation is performed [5]. The 10 fold cross validation mixes the data set randomly and divides it into 10 equally sized disjunctive partitions. Each partition is then used once for testing, the other partitions are used for training. This results in 10 different accuracies (the percent rations of the number of the samples classified correctly and the total number of testing samples), which are averaged.

Figure 4.3 shows that morphological method is highest accuracy in narrow bands. This fact makes it easy to find out the significant band. On the contrary, result of other method does not change as to be able to specify the significant band. It seems that this is caused by the fact that the high frequency information influences the higher levels. This phenomenon is most remarkable on the C Spline wavelet method for which the shape of mother wavelet is the sharpest among them. On the other hand, since morphological method is composed of max-min operations, high frequency information doesn't influence the higher levels. Therefore, morphological method is the most suitable tool from the view point of feature extraction.

In Figure 0.1, we show the learning effects of subjects for a 10 days experiment.

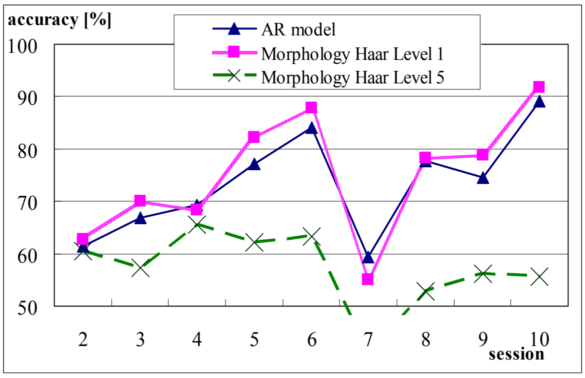


Figure. 4.4 Learning effect (channel: C3, subj.: A)

By a method of pattern recognition with AR-model it can be observed that the learning effects of subjects improve as they experience the experiments. But, in case of pattern recognition with IV feature, learning effects in different levels show different aspects. In learning using level 1 signal, the tendency to look like the AR method was obtained. On the contrary, for level 5 signal, the accuracy of identification is apparently low.

## 5. Conclusion

In this work, we applied wavelet and morphological multiresolution analysis on EEG signals during right and left hand motor imagery for feature extraction and pattern recognition. Intertrial Variance, extracted by multiresolution analysis, enables us to confirm changes of EEG caused by motion imageries in various frequency components simultaneously.

As an experimental result, components according to ERD/ERS in a certain bandwidth (levels 1-4) were observed and by taking their weighted distances from the average wave as features, a high accuracy of identification was achieved. In particular, since the feature extracted by the mathematical morphological method appears in a narrow bandwidth (levels 1-2), the specific band (levels 1-2) have an essential information about motor imagery. Furthermore, by training of subjects for this experimental system, the accuracy of identification in the bandwidth of low levels (high-frequency components) were observed tendency similar to AR method. This seems us to AR-model to feature the high-frequency components of the modeling. To verify this, we will construct a feedback system using the weighted distance as feature, and compare it with systems using AR-model as feature in the future work. Also, we need further analysis on EEG signals in each band.

## References

[1] J. J. Vidal, "Toward direct brain computer communication," Annual Rev. Biophysics Bioengineering. pp. 157-180, 1973.

[2] J. Wolpaw and D. McFarland, "Multichannel EEG based brain computer communications," Electroencephalography and clinical Neuro-physiology. 90, pp. 444-449, 1997.

[3] G. Pfurtscheller, Ch. Neuper, D. Flotzinger and M. Pregenzer, "EEG-based discrimination between imagination of right and left hand movement," Electroencephalography and clinical Neurophysiology, 103, pp. 642-651, 1997.

[4] C. Guger, "P300 Spelling Device with g.MOBIlab and Simulink", http://www.gtec.at/service/Tutorials/P300SpellerwithgMOBIlab.pdf

[5] Katsuhiro Inoue, et al, "Pattern Recognition of EEG Signals During Right and Left Motor Imagery Learning Effects of the Subjects ", edited by J.L Wu, K. Ito, S. Tobimatsu, T. Nishida, and H. Fukayama, Complex Medical Engineering, pp.251-261, Springer 2007

[6] C. Chui , "An Introduction to Wavelets," Boston, Academic Press, 1992

[7] J. Serra, "Image Analysis and Mathematical Morphology", London, U.K. Academic, 1994.

[8] H. J. A. M. Heijmans, J. Goutsias et. al. "Nonlinear Multiresolution Signal Decomposition Schemes-Part II:Morphological Wavelets," IEEE Trans. on Image Processing Vol. 9 No. 11, 2000.

[9] Fernando Lopes, M.D., Ph.D. Da Silva, F. H. Lopes Da Silva, "Electroencephalography: Basic Principles, Clinical Applications, and Related Fields," Lippincott Williams & Wilkins